



拉合尔供水与污水处理项目（LWWMP）
AIIB 项目编号 P000074 合同编号 0074-PAK-01

第三版澄清文件

单阶段两信封工程采购模式（EPC模式/交钥匙方式）

拉合尔污水与排水管理项目（LWDMP）——从LARECHS COLONY至GULSHAN E RAVI的

拉合尔污水系统（采用非开挖技术）

2025年12月17日



ASIAN INFRA
INVESTMENT

投标人疑问的第三版澄清文件

Sr #	Reference Section	Description	Employer's Response
1.	Volume 1, Part I, Section 4, Tender Forms Page 4-46-4-62 Schedule No. 5: Electrical Works	Electrical works in 23 Intermittent Pumping Stations Electrical works along sewer lines (15 Nos. Control System at Drop Structures) & 15 Nos. Control Points on Manholes along Lines and Laterals preferably at drop structures in accordance with ER Sub Clause 4.1 & 5.15 Nos. (minimum) Control Points on Manholes along Lines and Laterals ULTIMATE DISPOSAL STATION AT GULSHAN E RAVI (ELECTRICAL WORKS FOR PUMPING STATION AND ALL BUILDINGS REFERRED IN EMPLOYER'S REQUIREMENTS The Tenderer notes that only the Single Line Diagram (SLD) for the Disposal Station has been provided. The Tenderer respectfully requests the Employer to provide the load list, SLDs, layout drawings, and any additional electrical drawings for other stations, if available.	It is to clarify that EPC Contractor is responsible for detailed design of loads and SLDs for all the stations, however, the load requirement for all 23 intermittent pumping stations and typical SLDs for odor control system, ventilation system and screening system is attached as Annex-A for guidance only.
2.	Volume 2 Part II, Section 6, Employer's Requirements	It is recommended that the panel rooms can be adjusted in the dry well chamber. Can this alternate arrangement be acceptable to the Employer.	Kindly refer to response at Sr. No. 39 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, wherein, the updated drawing # LW-TD-411 shall prevail, however, as an alternative arrangement, the panel rooms can be adjusted in the dry well chamber as per Annex-B.
3.	Clarifications and Minutes of Meeting of 1st Pre Tender Meeting LWDMP Item 72 Volume-1, Part I, Price Schedules, Schedule No. 5: Electrical Works, Item No. 5.6.	It is to clarify that shifting of existing Generators and Control rooms at 23 lift stations at new location (within the existing premises) will require Medium Voltage Cables for restoring / reinstating the connection with the existing Pumping system. In accordance with the Employer's requirements for Medium Voltage (MV) cables to restore the connection with the existing pumping system, the Tenderer requests clarification on the following: 1) Please provide the load information, detailed load	1) The EPC contractor is at liberty to visit the relevant site and estimate the relocation and reconstruction scope of work, as per its own design and layout. Further, refer to response of Item 72 of Clarifications and Minutes of Meeting of 1st Pre Tender Meeting LWDMP, it is to clarify that 23 pumping stations other than Gulshan e Ravi pumping station have LV loads and these pumping stations will require Low Voltage Cables for restoring / reinstating the connection with the existing Pumping system, if shifting involves.

Sr #	Reference Section	Description	Employer's Response
		lists, Single Line Diagrams (SLDs) for MV pumps at each station, and all relevant interface details. 2) Apart from the MV motors, please clarify whether any existing facilities at the 23 lift stations require power supply from the relocated panel room (substation). If the design, procurement, and installation of power cables and distribution feeders fall within the Contractor's scope, please provide the corresponding load information, SLDs, and interface details for each station.	2) The EPC contractor is at liberty to visit the relevant site and estimate the relocation and reconstruction scope of work, as per its own design and layout, however, the equipment list of 23 pumping stations is attached as Annex-C for reference only. Since the relocation shall be planned within the existing premises, the same power supply source may be utilized for the re-located facility. It may be noted that due to space constraint in some of the disposal stations, the re-location job may also involve shifting of main supply source like transformer / generator, which may also involve MV cable,
4.	Page 6-20 of Volume 2 Part II, Section 6, Employer's Requirements Item 4 of Provisional Sums page 4-75 of Volume 1, Part I, Section 4, Tender Forms Schedule No. 5: Electrical Work, ULTIMATE DISPOSAL STATION AT GULSHAN E RAVI	Page 6-20 of V2 - Part II - Employer's Requirements. C. Electrical, Instrumentation & Control (EI&C) Design Assistance in Design of Grid Station with LESCO: Design of electricity 132 kVA grid station near ultimate pumping station and coordinate with LESCO. As per Schedule No. 5: Electrical Works (5.19-5.49) of V1 - Part I - Tender Procedures, the scope of the 132 kV GIS, 132/11 kV power transformer, and associated equipment installed at the 132 kV station has not been specified. The Tenderer understands that the 132/11 kV grid station is outside the Contractor's scope and that the battery limit terminates at the incoming switchgear of the 11 kV switch room, as shown in the Ultimate Disposal Station SLD. Kindly confirm this understanding.	Refer to Section 4, Tender Forms, Provisional Sums and Schedule No. 5: Electrical Work, Ultimate Disposal Station at Gulshan e Ravi, it is clarified that the grid station is also within the EPC Contractor scope, however, the detailed design of the grid station shall be prepared in accordance with the LESCO / WAPDA specifications and the EPC Contractor will coordinate with them for the installation of the grid station and its connection with the proposed pumping station at Gulshan e Ravi. The battery limit of the pumping station includes the grid station as provided in the provisional sums.
5.	Volume 1, Part I, Section	(23 Nos. Intake Structures for 23 Screen Systems &	The list of 14 generators with their rating and capacity is

Sr #	Reference Section	Description	Employer's Response
	4, Tender Forms Page 4-46 Schedule No. 5: Electrical Works Volume 2, Part II, Section 6, Employer's Requirement	16 control rooms & 28 buildings/ rooms reconstructed & 14 generators operated & 11 control panels shifted approximately in accordance with ER Sub Clause 4.1 & 5.1). The Tenderer requests the Employer to provide details for the 14 generators, including voltage rating, capacity, dimensions, and any additional relevant information. Further, please provide a basic design or overview of the Electrical (EL) System covering all stations under this project.	attached as Annex-C. The EPC contractor is at liberty to visit the relevant sites and prepare the design of proposed infrastructure within available premises by keeping existing system intact as per criteria / specifications given in Employer's Requirements. In addition, details regarding available load at 23 lift stations and other electrical installations is made part of Annex-C
6.	Volume 1, Part I, Section 4, Tender Forms Page 4-46 Schedule No. 5: Electrical Works Volume 2, Part II, Section 6, Employer's Requirement	(23 Nos. Intake Structures for 23 Screen Systems & 16 control rooms & 28 buildings/ rooms reconstructed & 14 generators operated & 11 control panels shifted approximately in accordance with ER Sub Clause 4.1 & 5.1). The Tenderer notes that only the 14 generators located at the existing intermittent pumping stations are mentioned in the ITB. Please clarify: 1) Whether the remaining 9 stations require new generators or if they have existing generators that may be utilized. 2) Whether the capacity of the 14 existing generators is sufficient to meet the new stations' load requirements.	It is to clarify that list of Generators available at 23 lift stations is attached at Annex-C, however, as per preliminary design and drawings, shifting of only 14 generators is envisaged, however, the Tenderer is at liberty to avoid shifting during formulation of its design / costing. Further, 1) As per current assessment, new generators are not required at remaining 9 intermittent pumping stations. 2) The capacity of the existing generators is considered to be sufficient to operate the attached loads.
7.	Volume 2, Part II, Section 6, Employer's Requirement xxxix) Design and supply of SCADA System, Page6-73	The SCADA hardware will include: Sewage video cameras and surveillance systems for visual inspection of trunk sewerage lines and disposal stations. According to the referenced clause, please clarify whether video surveillance cameras are required along the entire sewage pipeline route or only at critical facilities.	As responded in Sr. No. 90 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, it is to clarify that Sewage Video Cameras shall be installed at critical facilities. The cameras will be installed, where ever RTUs are to be installed in the manholes along the sewage pipeline and at critical locations as site requirements.
8.	Volume 2, Part II, Section 6, Employer's Requirement DATA ACQUISITION FOR SCADA AT VARIOUS	a) Remote Terminal Units (RTU) shall be installed at selected (jacking shafts) sites to retrieve the data and transmit through the 4G (GSM/GPRS) media to the new disposal station at Gulshan Ravi. List of RTU sites is provided.	It is to confirm that all RTUs and cameras at pipeline shall communicate using 4G media.

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	SEWERAGE COLLECTION ON MAIN SEWER LINES, Page 6-332	b) Option of wireless radio is recommended as primary media as it is readily installable and cover wide area of communication. Data transmitted per station, in general for this project, is in the range of 1 ~ 5, which is easily manageable with 4G technology with required performance. As per the clause and Volume 2, Part II, Section 6 of the Employer's Requirements (Page 6-323), the Tenderer understands that all RTUs and cameras at pipeline stations will use a leased 4G GSM wireless network for data transmission, and that no new optical fibre cables will be installed. Please confirm this understanding.	
9.	Volume 2 - Part II - Employer's Requirements Sub-Clause 4.1 Contractor's General Obligations and Volume 3-A - Part II -Tender Drawings	Line-A will collect flows from disposal stations of Karachi Phattak/ Harbanspura DS, Main Mir Colony, Infantry Road, Bird Market, Larech Colony, Ghazi Mohalla, Muhammad Nagar, Governor House, Lakshmi Chowk, and Janki Devi. Line-B collect sewage flows from disposal stations Guru Mangat, Q-Block, CentrePoint, G-Block, Zafar Ali Road, Shareef Colony, Ichra, Shadman, Shama, and Rasool Park. Line-C collects sewage flows from disposal stations of Lytton Road, PIA Planetarium, and Poonch Road. The Tenderer notes that no drawings have been provided in V3A - Part II - Tender Drawings for the Harbanspura / Karachi Phattak Disposal Station. Kindly define the scope of works and provide drawings for the inlet chamber, outlet chamber, mechanical screen, drop structure, and odour-control system	Please refer to the reply to Sr. No. 26 of the Clarifications and Minutes of Meeting of 2nd Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.
10.	Volume 3-A - Part II - Tender Drawings Drawing no: LW-TD -491 ULTIMATE DISPOSAL STATION SINGLE LINE DIAGRAM (SLD)	In Drawing No: LW-TD -491, there are water supply pumps and fire pumps however, the relevant parameters and requirements for these facilities are not specified in the tender documents. The Tenderer requests clarification on the following: 1) Whether fire water supply facilities are required at the Ultimate Disposal Station. If yes,	1. Kindly refer to Section 6 Employer's Requirements, Environmental & Social Management Plan (ESMP), Table 1-5: Environmental and Social Management Plan during Construction and Operational Phase, it is to clarify that fire water supply facilities are required at the Ultimate Disposal Station and the system will be designed and constructed according to National Fire Protection Association (NFPA)

Sr #	Reference Section	Description	Employer's Response
		please provide the type and design parameters (flow rate, head) of the fire pumps. 2) Please specify the fire-fighting systems required for each building (fire hydrant system, gas-based fire-extinguishing, automatic sprinklers, etc.).	standards, by using design., codes/references mention hereunder: NFPA 20 (fire pumps), NFPA 13 (automatic sprinklers), NFPA 14/NFPA 24 (hydrants / private mains), NFPA 22 (fire water tanks), and NFPA 2001 (clean-agent systems). Provide a standalone fire water supply capable of supporting private hydrants and sprinklers for a minimum 2-hour duration dedicated water tank at the chosen design flow. System Requirement: Install external hydrants (industrial style) distributed to provide required coverage; hydrant outlets, 2.5-in nozzles, and provisions for fire department connections (FDC). Plant buildings, workshops, stores, hazardous material storage (if any): Hydrant system inside/outside + hose reels as required; consider local sprinklers inside enclosed buildings where combustible materials are stored. Additional: manual fire extinguishers, detection & alarms, emergency lighting, and signage. Jockey pump (pressure maintenance): Jockey pumps are small, typical flow 5-10% of main pump flow to maintain pressure and avoid main pump cycling. Design Pressure: Consider a minimum of 7 barG at the hydraulically most remote hydrant/nozzle. The pump head shall be calculated based on the terminal pressure of 7 barG + static head + friction losses. Design Flows: EPC contractor shall calculate the fire pump flow rate considering the worst-case scenario, but shall not be less than 1000 gpm. Fire Pump shall be FM-approved and UL-listed. Large Electric Motors: In case of a large pump room, partially open and containing lubricants/oils, install both local deluge zones (for fuel fires) and gaseous total-flooding for electrical enclosures. Systems must be zoned to avoid water flooding live switchgear. 2. For the offices at the Ultimate Disposal Station, the Contractor shall design and provide an Automatic wet-pipe sprinkler system (light hazard).

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		Also clarify whether a fire water tank is required and, if so, the required capacity. 3) Whether gas-based fire-extinguishing systems are required in specific rooms such as the control room or substation. If so, kindly identify the applicable rooms. 4) Whether a potable water supply system is required. If yes, please provide pump parameters, tank size, and the source of domestic water. 5) Whether integrated sewage treatment facilities are required at the station and clarify the method for domestic sewage disposal.	3. For the electrical building and pump room, where electrical equipment are installed, at the Ultimate Disposal Station, the Contractor shall design and provide a clean gas-based fire suppression system e.g. Inergen (IG-541) or FM200. For the 23 intake locations, the Contractor shall provide powder-based fire extinguishers. 4. The Contractor shall ensure the supply of potable water at all intake locations and at the ultimate disposal station, including the new building. The primary source shall be the WASA-L water supply, if available in the vicinity. If WASA-L supply is not available, groundwater shall be used as the alternative source, and the Contractor shall install a borehole pump with all associated works. The Contractor shall design the water supply system for the mentioned locations in accordance with WASA-L criteria. 5. There is no requirement to treat the domestic sewage before disposal into the sewerage system at the intake locations and the ultimate disposal station.
11.	Volume 2, Part II, Section 6, Employer's Requirement Sub-Clause 5.1 General Design Obligations	Penstock Gate Valves Operation method: Manual, rising stem in 23 lift station and ultimate disposal station. Please clarify whether motorized actuators are required for Penstock Gate Valves. If required, kindly specify which Penstock Gate Valves in the 23 lift stations and the Ultimate Disposal Station are to be equipped with motor-driven mechanisms.	All penstocks shall be provided with hand-operated gearboxes, designed to allow full opening and closing by a single operator under all operating conditions.
12.	Volume 2, Part II, Section 6, Employer's Requirement 4.1.12. Detailed Description of Scope of Construction Works by the EPC Contractor	xx) Construction of Office Building (Drawing No. LW-TD-401 through Drawing No. LW-TD-403) Page 6-67. Please provide the quantities and specifications of all facilities to be provided within the Office Building.	Please refer to the reply to Sr. No. 44 and 45 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.

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13.	Volume 2, Part II, Section 6, Employer's Requirement Sub-Clause 6.6 Facilities for Staff and Labor Page 6-397	Pursuant to Sub-Clause 6.6 of the Conditions of Contract, the facilities for Employer's Person to be provided by the Contractor are mentioned as follow: The Contractor shall ensure following facilities for Employer: • Fully furnished & air-conditioned site offices along with retiring room and all office amenities/ facilities for Employer & PMC Site Staff. • Arrangement of food & drinks at offices • Provide proper soft & hard record maintenance arrangements • Design Office with Committee / Meeting Room equipped with multimedia wall and interactive 3-D model of project for demonstration. • Availability of all stationery, colored & black & white printing & scanning arrangement. • Day care arrangement for Staff with kids. • Office area to be fully secured with security facility, parking area, Camera Surveillances etc. • Provide latest Office equipment to Employer's staff (i.e. Office computers/ laptops with licensed software & windows, and mobile phones & tabs to record site activities & complaints or track progress of work etc.) • Provide reasonable transportation arrangement for the Employers and PMC Staff. • Make arrangement for staff accommodation near sites to facilitate site supervision & inspections. • Make arrangements for routine & high-level delegation inspections of work including earth-breaking & inauguration event. • Contractor shall setup Media handling arrangements for regular feeds to media. • Make arrangement for staff accommodation near sites to facilitate site supervision & inspections. • Make arrangements for routine & high-level delegation inspections of work including earth-	Pursuant to Sub-Clause 6.6 of the Conditions of Contract, the Contractor shall ensure site offices preferably at each active site, in no case the minimum number of site offices shall be less than three (03) at the designated site locations. The containers shall be separate for Employer, PMC and Contractor etc. use. These offices shall be provided in accordance with the facilities, specifications, and standards stipulated in the Employer's Requirements (ER).

Sr #	Reference Section	Description	Employer's Response
		breaking & inauguration event. • Facilitate preparation of record and evidences for accountability agencies. • The Contractor shall bear all such costs to be included in bid price. Please specify the number and type of offices to be provided for the Employer, along with the required facilities and specifications.	
14.	Volume 2, Part II, Section 6, Employer's Requirement iii) Component 1: Laying of Trunk Sewers Through Micro-tunneling, Construction of Sewer Appurtenant Structures, and Installation of E&M Equipment Page 6-28	The Contractor shall ensure Manufacturing, Procurement, Delivery and Deployment of at least six number of (06) New MTBMs (OEM Certified) at site with all accessories/ equipment's including but not limited to RT Cranes, dump truck trolleys, trailer, excavator and their power supplying generators with trained O&M professionals and adequate spares readily available at site. The MTBM Machines are recommended to be available and operating at site before 6-months of the commencement of work to ensure timely completion. In no case it shall be delayed more than 8 months for MTBM. These Machines shall be capable to complete the project within the project time without major overhauls during the construction period: Diameter 1.050 m: Two (2) machines Diameter 1.500 m: One (1) machine Diameter 2.400 m: Two (2) machines Diameter 3.500 m: One (1) machine According to the clause requiring the Contractor to procure six new MTBM machines, please clarify the ownership of this equipment upon completion of the Project.	The MTBM machines required for execution of the said Works shall remain the property of the Contractor upon successful completion of the Project.
15.	Volume 2, Part II, Section 6, Employer's Requirement, 4.1.3. THE PROJECT BRIEF Page 6-6	Sewage conveyance /transport system consists of Line-A, Line-B, Line-C and their laterals which consist of about 19.0 km long Trunk Sewers and 13.36 km Lateral/branch and will be laid by micro-tunneling and pipe jacking technique with varying diameter of pipes from 1,050 mm, 1,500 mm, 2,400	Please refer to the reply to Sr. No. 09 of the Clarifications and Minutes of Meeting of 2 nd Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.

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		mm and 3,500 mm. Based on the Employer's Requirements and Drawings, the 2400 mm and 3500 mm diameter pipelines are to be installed using micro-tunnelling and pipe-jacking. Please clarify whether the Tenderer may propose an alternative trenchless technology in lieu of the specified method.	
16.	Volume 2, Part II, Section 6, Employer's Requirement	Please clarify the Employer's position regarding the sourcing of major construction materials such as cement and steel, particularly in relation to locally available materials versus imported materials.	It is to clarify that the sourcing of construction materials shall remain the Contractor's responsibility, subject to compliance with the technical specifications and applicable laws and regulations and Employer's consent. In this regard, Contractors are advised to carefully consider the time-related and fiscal implications of importing materials, including applicable taxes, duties, and other related charges. All commonly available and locally manufactured materials forming part of the Permanent Works, including but not limited to cement, steel, sand, aggregates, and similar construction materials, are required to be procured locally from within Pakistan. Import of such locally available permanent materials shall not be permitted and shall require prior written approval of the Employer, supported by adequate technical justification. It is further clarified that the Employer's liability for payment of taxes and duties shall be limited to permanent plant and equipment (machinery) of Employer's use to be installed under the Project. Such taxes and duties shall be paid by the Contractor at the time of import and shall be reimbursed by the Employer through the Interim Payment Certificates (IPCs), in accordance with the provisions of the Contract.
17.	Volume 2, Part II, Section 6, Employer's Requirement	During the site visit, the Tenderer has identified an additional parcel of land probably owned by Lahore Development Authority (LDA), located in proximity to the Project area and WASA owned land, which appears suitable for use as a pipe manufacturing and casting yard. In this regard, the Tenderer seeks clarification	It is to clarify that in addition to the sites identified in the Tender Documents, the Contractor may utilize any other suitable land, including the land owned by Lahore Development Authority (LDA) identified during the site visit, for pipe manufacturing and associated activities, subject to obtaining all requisite approvals and NOCs from the relevant authorities. In this regard, the updated

Sr #	Reference Section	Description	Employer's Response
		whether, subject to obtaining all requisite approvals and permissions from the relevant authorities, the said land may also be utilized for pipe manufacturing, storage, and associated construction activities under the Contract.	drawing for sites that can be used by the Contractor are attached as Annex-D.

附件A

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Drop Structure Odor Control System						
Sr.No.	Location	Installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir (D1)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
2	Infantry (D2)	4	1	DOL	MCB -TP, 10A-16kA	MCCB-TP, 20A - 25kA
3	Bird Market (D3)	7.5	1	DOL	MCB -TP, 20A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
4	Gov House (D4)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
5	Lakshami Chowk (D5)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
6	Line-A Qurtaba (D6)	15	2	DOL	MCB -TP, 40A-16kA	MCCB-TP, 50A - 25kA
7	Grumangat(D7)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
8	Q Block (D8)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
9	G Block (D10)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		2.2	1	DOL	MCB -TP, 6A-16kA	
10	Zafar Ali Rd (D11)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
11	Ichra (D12)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
12	Shama (D14)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
13	Line-B Qurtaba (D16)	11	1	ASD	MCB -TP, 32A-16kA	MCCB-TP, 50A - 25kA
		7.5	1	DOL	MCB -TP, 20A-16kA	
14	Lyton (D17)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
15	PIA (D18)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
16	Poonch (D19)	4	1	DOL	MCB -TP, 16-16kA	MCCB-TP, 20A - 25kA

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Screens Area						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
2	infantry	3	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 16A-16kA
3	Bird Market	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
4	Larech Colony	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
5	Ghazi Mohallah	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
6	Mohammad Nagar	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
7	Lakshami Chowk	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Lakshmi Chowk Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
8	Gov House	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
9	Janki Devi	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
10	Grumangat	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
11	Q Block	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
12	Center Point	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
13	G Block	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
14	Shareef Colony	0.55	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
15	Zafar Ali Rd	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
16	Shadman	2.2	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
17	Shama	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
18	Ichra	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
19	Rasool Park	2.2	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Rasool Park Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
20	Lyton /Lake Rd	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
21	PIA	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
22	Poonch	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Poonch Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Ventillation System						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating (kW)	Installed Qty. (Nos.)	Starter Type	Feeder Circuit Breaker	
1	E1	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
2	E2	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
3	E3	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
4	E4	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
5	E5	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
6	E6	2.2	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
7	E7	3	1	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
8	E8	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
9	E9	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
10	E10	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
11	E11	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
12	E12	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
13	E13	15	1	ASD	MCB -TP, 40A-16kA	MCB -TP, 63A-16kA

附件B

Description	Area (m ²)
Total Area available at GR Disposal	17741.371
Area occupied by Old Disposal	1600.805
Area Occupied by New Disposal including DC	3376.834
Admin Building (offices)	296.038
Remaining area in pieces (After all construction)	12467.69

附件C

Sr	No	Name of Lift Station	Number & Capacity of Pumps	Transformer	Generator	Remarks
(Motor Rating)	(KVA)	(KVA)				
1	Mian Mir	1	No x 4 Cusec (60 HP)	100	Not Available	
1	No x 4 Cusec (50 HP)					
2	Infantry Road	1	No x 4 Cusec (50 HP)	200	200	Shifting of Generator
1	No x 6 Cusec (50 HP)					Requires as per PMC Design
1	No x 10 Cusec (75 HP)					
1	No x 10 Cusec (100 HP)					Shifting of Generator
3	Bird Market	1	No x 10 Cusec (100 HP)	400	300	Requires as per PMC Design
1	No x 25 Cusec (200 HP)					
	Larech Colony	1	No x 2 Cusec (20 HP)	100	200	Shifting of Generator
4	1	No x 6 Cusec (60 HP)				Requires as per PMC Design
5	Ghazi Mohalla	1	No x 4 Cusec (30 HP)	100	200	Shifting of Generator
1	No x 6 Cusec (60 HP)					Requires as per PMC Design
6	Muhammad Nagar	1	No x 6 Cusec (80 HP)	50	200	
1	No x 6 Cusec (60 HP)					
7	Governor House	1	No x 6 Cusec (60 HP)	2 x 200	KVA	Not Available
1	No x 6 Cusec (60 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 25 Cusec (215 HP)					500 KVA + 500 KVA + Shifting of Generator
8	Lakshmi Chowk	1	No x 25 Cusec (215 HP)	400 & 1000	KVA 1000 KVA	Requires as per PMC Design
1	No x 25 Cusec (215 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 6 Cusec (60 HP)					Shifting of Generator
9	Lytton Road	1	No x 10 Cusec (80 HP)	200	KVA 200 KVA	Requires as per PMC Design
1	No x 6 Cusec (40 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 12 Cusec (100 HP)					
10	PIA Planterium	1	No x 12 Cusec (100 HP)	400 & 200	KVA 200 KVA & 1000 KVA	
1	No x 12 Cusec (100 HP)					
1	No x 25 Cusec (215 HP)					
1	No x 25 Cusec (150 HP)					
11	Gurumangat Road	1	No x 15 Cusec (150 HP)	200	KVA 500 KVA	
1	No x 15 Cusec (150 HP)					
12	Q Block Gulberg	1	No x 25 Cusec (150 HP)	200	KVA 200 KVA	
1	No x 6 Cusec (50 HP)					
1	No x 25 Cusec (150 HP)					
1	No x 25 Cusec (150 HP)					Shifting of Generator
13	G Block Gulberg	1	No x 25 Cusec (150 HP)	1000	KVA 500 KVA	Requires as per PMC Design
1	No x 25 Cusec (150 HP)					
1	No x 25 Cusec (150 HP)					
14	Zafar Ali Road	1	No x 6 Cusec (50 HP)	100	KVA 200 KVA	Shifting of Generator
1	No x 6 Cusec (40 HP)					Requires as per PMC Design
15	Shareef Colony	1	No x 6 Cusec (40 HP)	50	KVA 200 KVA	Shifting of Generator
1	No x 6 Cusec (40 HP)					Requires as per PMC Design
1	No x 6 Cusec (40 HP)					
16	Shadman Colony	1	No x 25 Cusec (150 HP)	2 x 630	KVA 630 KVA	
1	No x 25 Cusec (150 HP)					
1	No x 10 Cusec (75 HP)					Shifting of Generator
17	Shama	1	No x 6 Cusec (80 HP)	100	KVA 200 KVA	Requires as per PMC Design
1	No x 6 Cusec (60 HP)					
1	No x 6 Cusec (60 HP)					
18	Ichra	1	No x 6 Cusec (80 HP)	800	KVA 300 KVA	
1	No x 6 Cusec (80 HP)					
1	No x 25 Cusec (150 HP)					
1	No x 15 Cusec (110 HP)					200 KVA & 630 Shifting of Generator
19	Rasool Park	1	No x 08 Cusec (80 HP)	KVA 500	KVA	Requires as per PMC Design
1	No x 06 Cusec (60 HP)					
1	No x 06 Cusec (110 HP)					
1	No x 10 Cusec (80 HP)					Shifting of Generator
20	Poonch Road	1	No x 10 Cusec (80 HP)	400	KVA 400 KVA	Requires as per PMC Design
1	No x 20 Cusec (150 HP)					
21	Jhan ki Devi	1	No x 01 Cusec (20 HP)	50	KVA	Not Available
1	No x 08 Cusec (50 HP)					Shifting of Generator
22	Centre Point	1	No x 06 Cusec (50 HP)	400	KVA 300 KVA	Requires as per PMC Design
1	No x 06 Cusec (60 HP)					
	Under Construction					
	Harbanspura /Karachi	5	No x 25 Cusec (150 HP)	1000	KVA 1000 KVA	(Expected to be completed

附件D

